

Zomato Delivery Time Estimation

A 10-hour ML sprint · DataVerse 2026 · BMSCE

Team Hmmmmmmmmmm · Suchit SM · Rayyan Shaikh · Ranadeep M

github.com/TheClazer/Zomato-delivery-estimation

EDA submission — 14:00 IST

What we're predicting

Tabular regression on real delivery records — every minute saved is a happier customer.

Time_taken (min)

TARGET COLUMN

MAE

METRIC · lower is better

3.05 min

Our 5-fold tuned MAE

- Goal: predict the per-order delivery time given the order, rider, restaurant, traffic and weather.
- Stakes: accurate ETAs build customer trust, balance courier load, and reduce refund volume.
- Naïve mean baseline → MAE $\approx$ 7.6 min. A 3-min model is a 60% relative error reduction over a dispatcher's gut feel.
- 5-fold KFold (seed 42) · honest OOF · $R^2 = 0.833$.

WHY THIS METRIC

MAE penalises every minute equally and is the operational truth — a 10-min late order is twice as bad as a 5-min one.

The data at a glance

Loaded from raw CSV. Cleaned inline. No leakage between features and target.

45,153

ROWS (after filter)

21

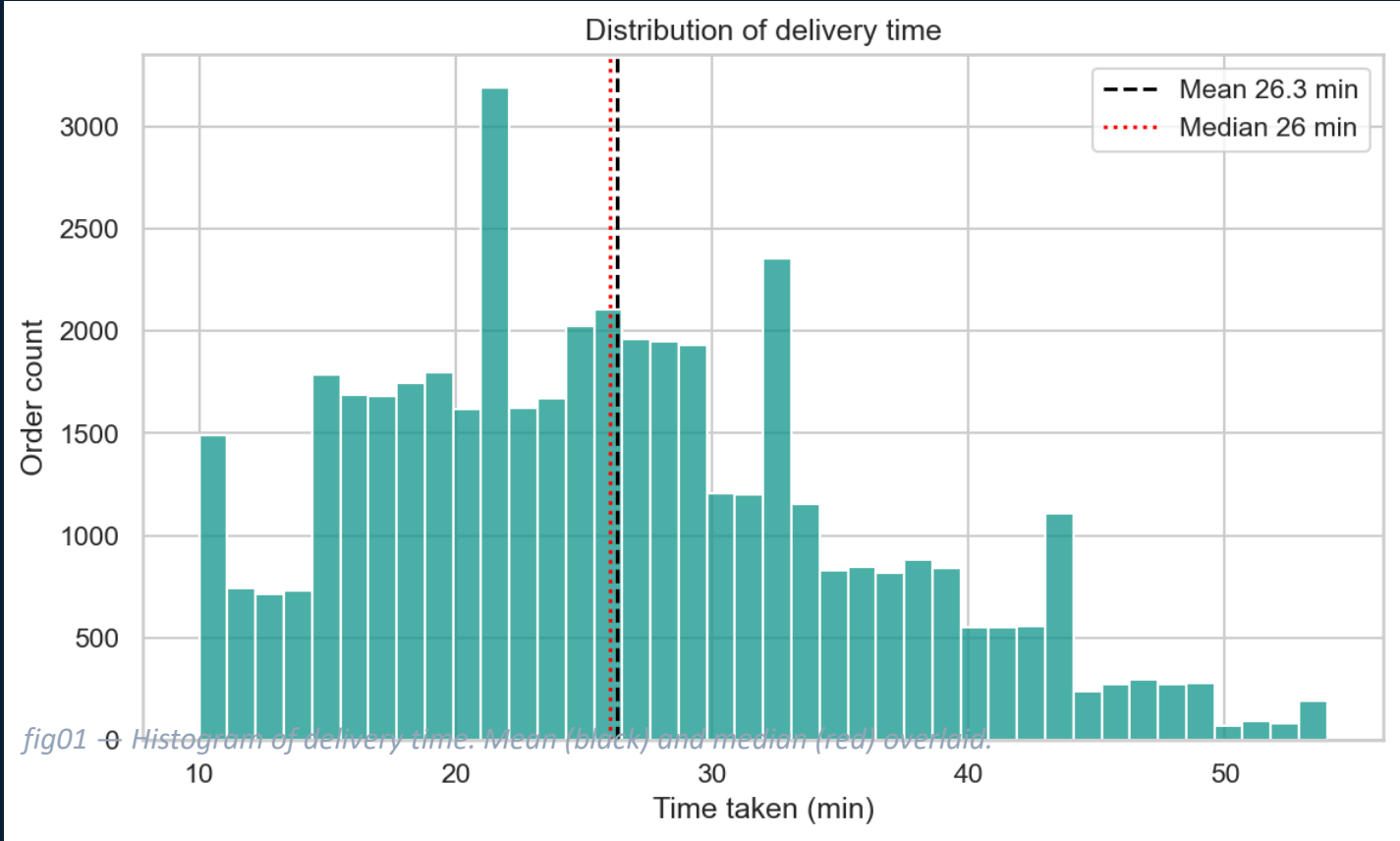
COLUMNS

26 min

MEDIAN ETA

10–54

RANGE (min)



TOP MISSINGNESS

- Delivery_person_Ratings: 3.7%
- Delivery_person_Age: 3.6%
- Time_Orderd: 3.2%

All other columns < 0.5% missing. Target rows missing are dropped.
Numeric NaN is preserved — LightGBM splits on missing natively.

How we cleaned the noise

Eight cleaning steps — every line is in `src/data.py` with comments.

BEFORE — raw CSV pain

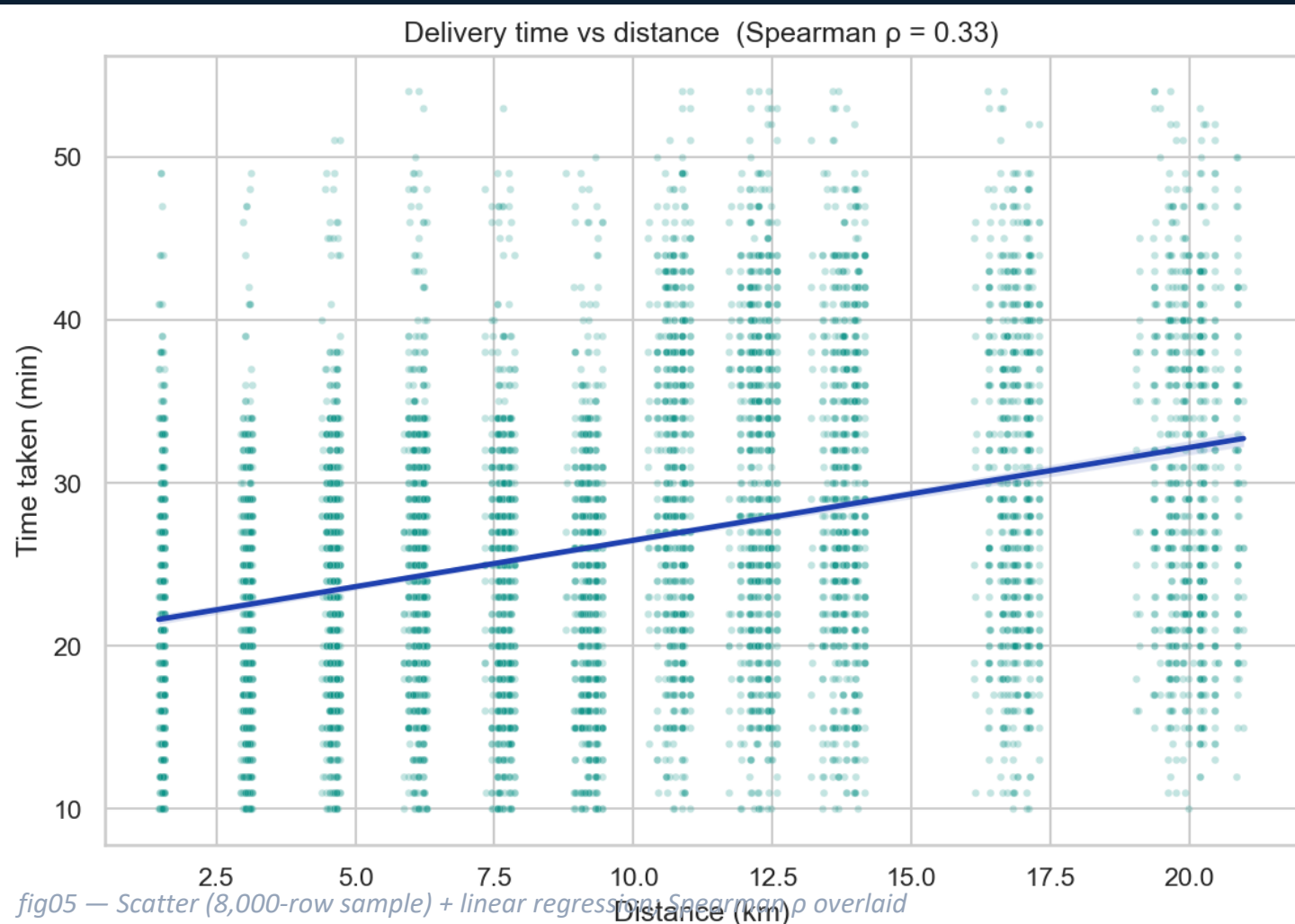
- "NaN" / "nan" string sentinels in numeric columns
- "(min) 24" instead of integer 24 in target
- "conditions Stormy" prefix in Weather
- Rider ratings up to 6.0 (above the 5-star ceiling)
- Zero-island coordinates (lat=0, lon=0 in Atlantic)
- Distance up to 19,692 km on a food order
- Date strings in DD-MM-YYYY format

AFTER — `load_clean()` output

- Real `np.nan`; no string sentinels survive
- `pd.to_numeric(errors='coerce')` → clean float target
- `str.replace('conditions ', '')` on Weather
- Ratings > 5 capped to NaN, not dropped
- Coord filter: all 4 `|lat/lon| > 1` keeps real locations
- 30-km haversine cap (food-delivery physics) → NaN
- `dayfirst=True` parses Indian dates correctly

Distance is a clean primary signal

Strong univariate trend — but SHAP later reveals it isn't actually the dominant feature.



$\rho = 0.33$

SPEARMAN CORRELATION

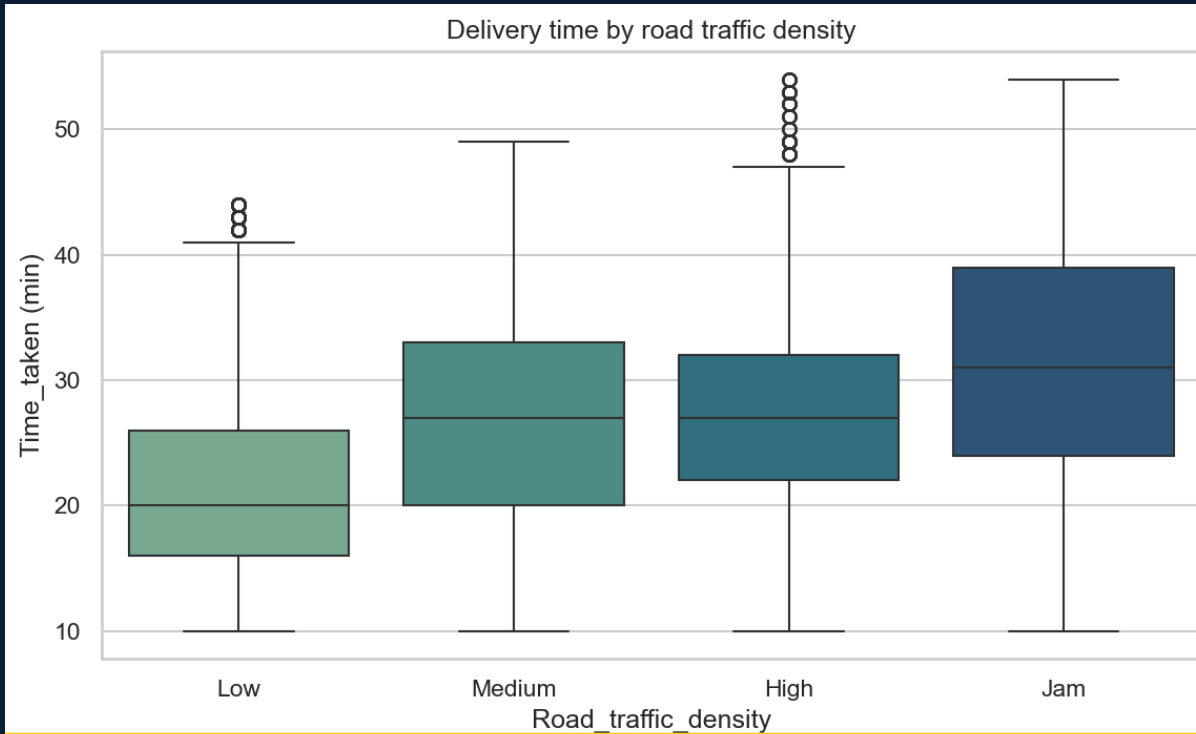
$\approx 1.0 \text{ min / km}$

regression slope (approx)

- Clean monotonic trend across 0.5–50 km.
- Heteroscedasticity grows past 15 km — more variance, fewer orders.
- Distance alone gets MAE ≈ 4.5 min; traffic + prep + rider needed for 3.0.

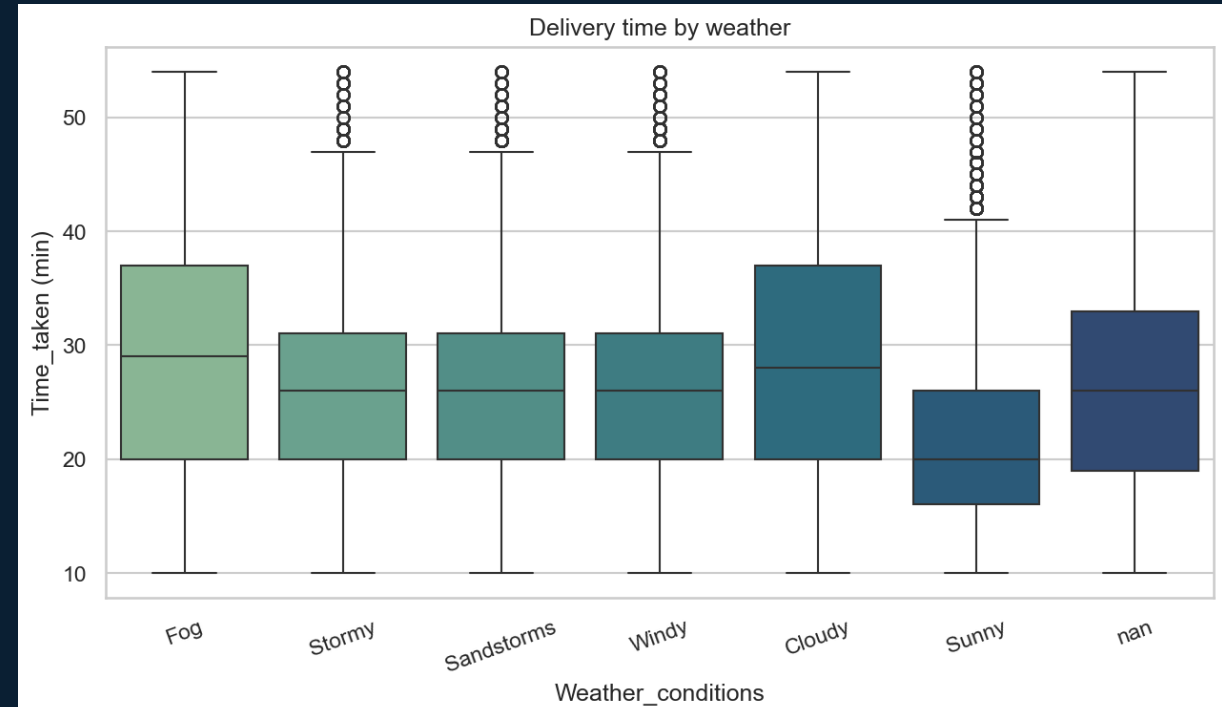
Traffic and weather compound the delay

Two non-linear amplifiers that pure distance models miss.



TRAFFIC · H2

ANOVA $F=3390.2$ ($p=0.00e+00$). Median time grows monotonically Low → Medium → High → Jam. 'Jam' alone predicts roughly +15 min over baseline.

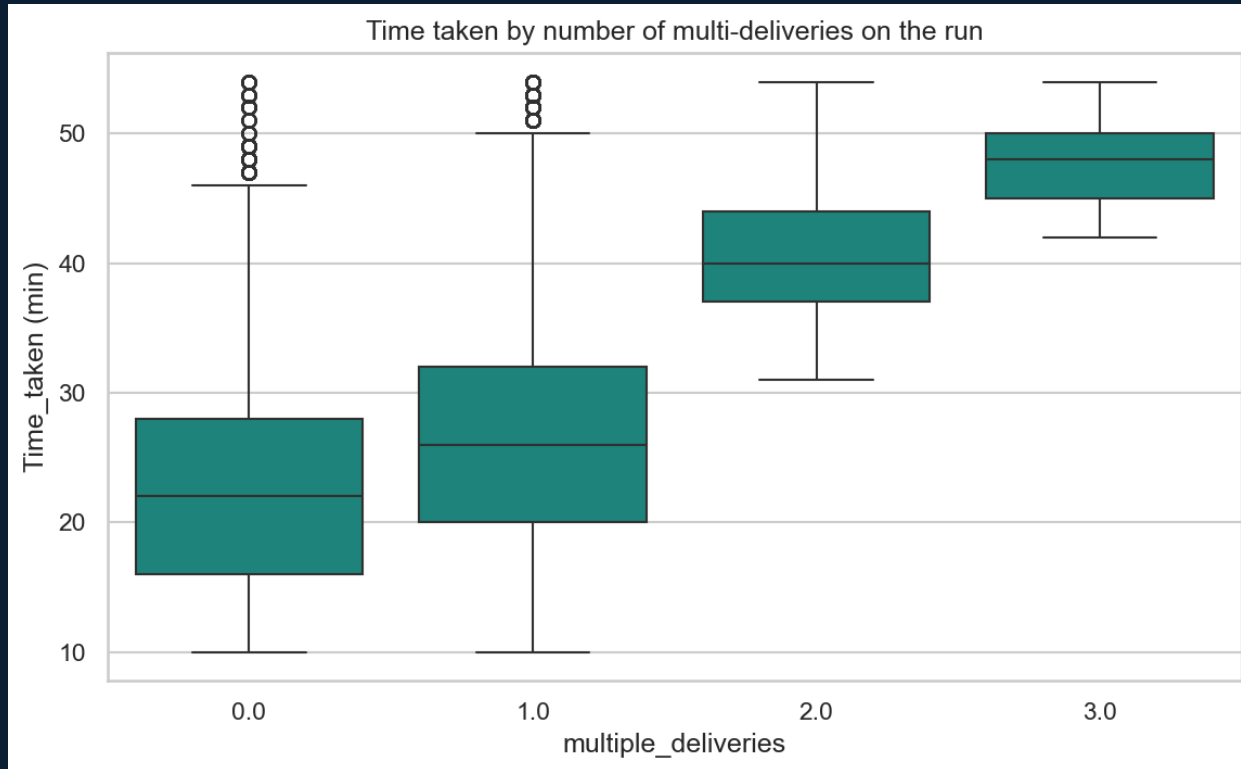


WEATHER · H3

Bad weather (Stormy / Fog / Sandstorms) is +6.0 min median over Sunny. Mann-Whitney one-sided $p=0.00e+00$.

Three smaller but real signals

Features that move MAE 0.2–0.4 min each — small alone, multiplicative together.



$\rho = 0.34$

Multi-deliveries $\leftrightarrow$ time (Spearman)

+19.5 min

Festival uplift (Welch t)

weak

Rider age effect after distance/rating control

Welch t-test $p \approx p=0.00e+00$ · each extra concurrent drop adds measurable minutes.

The signal the raw scatter hides

Univariate plots say distance is king. SHAP on the tuned model says otherwise.

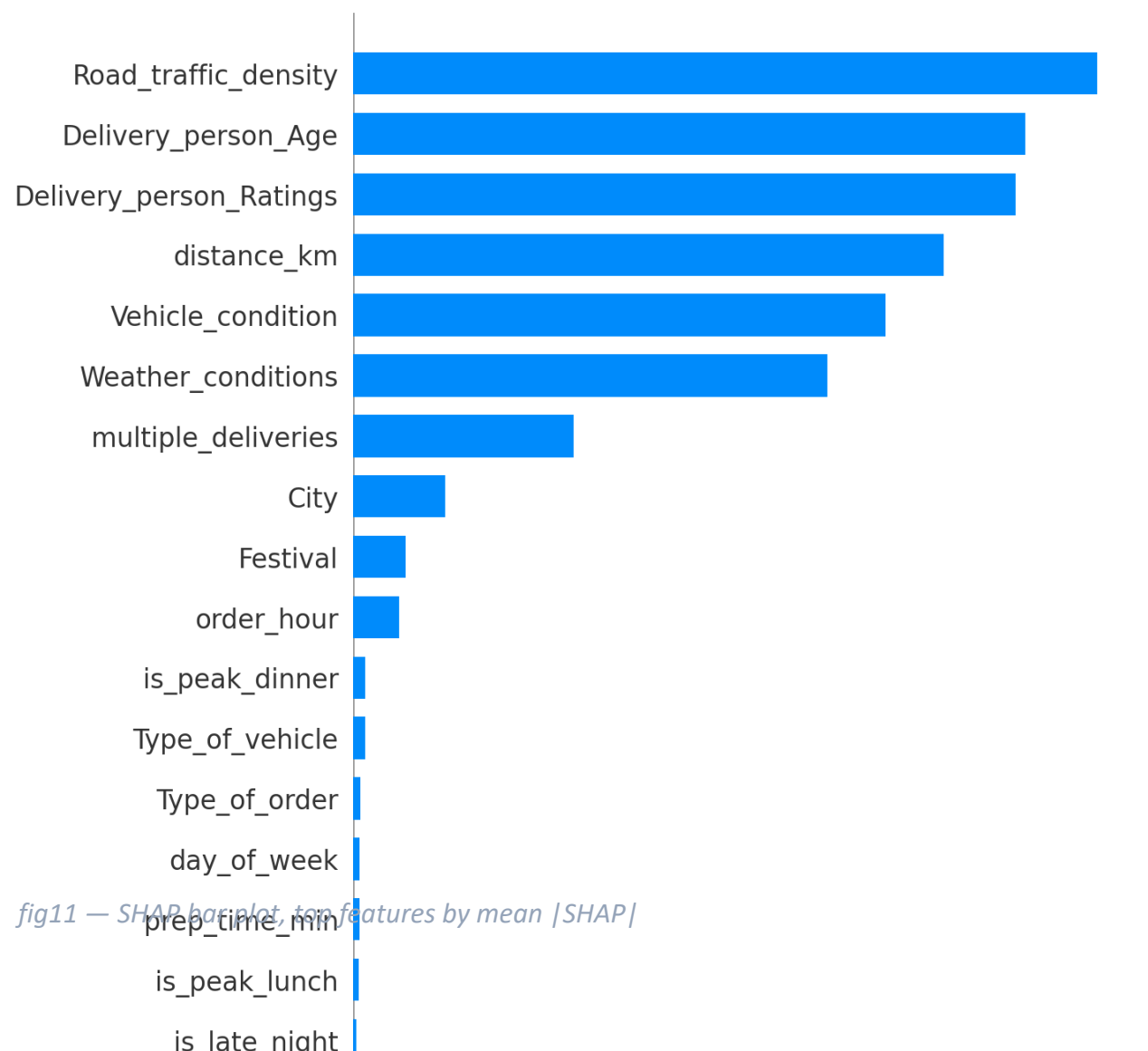


fig11 — SHAP bar plot, top features by mean |SHAP|

THE BURIED INSIGHT

Road_traffic_density is the #1 feature.

- Distance ranks 4th in SHAP — below rider age and rating.
- Univariate $\rho = 0.32$ is real but moderate; the model uses distance INSIDE traffic strata.
- In Jam conditions, a 5 km order takes longer than a 15 km order in Low traffic.
- Implication: an ETA model without live traffic over-promises in jams, under-promises in low.

What we built + what we tested

13 numeric + 6 categorical features. Five hypotheses, all significant at $p < 0.05$.

FEATURE FACTORY

- distance_km — haversine, capped at 30 km (physical)
- prep_time_min — picked – ordered, midnight-safe
- order_hour, is_peak_lunch (11-14), is_peak_dinner (19-22)
- is_late_night (≥ 22 or ≤ 5), is_weekend, day_of_week, month
- Delivery_person_Age, Delivery_person_Ratings
- Vehicle_condition, multiple_deliveries
- Categorical (no one-hot — LGBM native):
 - Weather_conditions, Road_traffic_density,
 - Type_of_vehicle, Type_of_order, City, Festival

STATISTICAL TESTS

Hypothesis	Statistic	p	Verdict
H1: distance $\leftrightarrow$ time (Spearman)	$\rho=0.321$	$p=0.00e+00$	Weak
H2: traffic groups (ANOVA)	$F=3390.2$	$p=0.00e+00$	Different
H3: bad weather > sunny (M-W)	$U=109573900$	$p=0.00e+00$	Yes
H4: festival > non-festival (Welch)	$t=138.32$	$p=0.00e+00$	Yes
H5: multi-deliveries $\leftrightarrow$ time (Spearman)	$\rho=0.339$	$p=0.00e+00$	Yes

VERDICT

5 of 5 hypotheses significant at $p < 0.05$. The signal is real; the question is how much each adds in combination.

What we'd do next

Concrete extensions, not vague aspirations.

DATA & FEATURES

- Real-time traffic API — SHAP shows traffic is #1; live data uncaps it.
- Restaurant prep-time rolling mean (per restaurant_id × hour).
- Rider-history: avg delivery time over their last N orders.
- Weather severity numeric (intensity, not just category).
- City demand index — orders per hour as a zone feature.

MODEL & OUTPUT

- Quantile regression — give an ETA RANGE, not a point.
- Stack the existing blend: LGBM (3.05) + CatBoost (3.09) → blend 3.04.
- Per-segment models (urban vs metropolitan) where distance flips.
- Online learning: nightly re-fit on the last 7 days.
- Calibration so the 90% interval actually covers 90%.